

Lecture 9: The Living Earth BTN

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

Contents

1	Lecture 9: the living Earth by the numbers	1
2	Age of the Earth – Darwin’s geomorphology	1
2.1	An updated estimate	2
3	Age of the Earth – Kelvin’s heat diffusion	3
4	Age of the Earth – from genomes	4
4.1	Poisson correction	5
4.2	Empirical mutation rate for proteins	6
4.3	Simple calculation	6
5	The size of the biosphere	7
5.1	Carbon mass of a single tree	7
5.2	Tree density from spacing	7
5.3	Putting it all together	7
6	The size of humanity	8
6.1	A generous over-estimate	8
6.2	A (somewhat) more detailed calculation	8

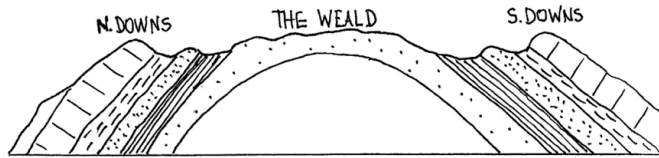


Figure 1: Stratigraphy of the North and South Downs.

Figure 1: An illustration of the Weald from “The Age of the Earth” by Archibald Hendry. The Weald is an area in South East England between the parallel chalk escarpments of the North and the South Downs. It is the eroded remains of a geological structure. See [Wikipedia](#) for more.

1 Lecture 9: the living Earth by the numbers

This planet is special. So far as we know, it's the only one with life. Certainly, it's the only one we live on.

If all life was gone, the rocks would keep on, but it would not be the same Earth.

Today we'll talk about life on Earth – thinking about deep time, evolution, how life has changed the planet, and the size of the biosphere.

2 Age of the Earth – Darwin's geomorphology

Evolution requires a lot of time. Darwin understood this from the very beginning. Since Darwin has studied geology, he attempted a geomorphological estimate as follows.

As you can appreciate from the excerpt below, Charles Darwin estimated the time to denude the Weald (Figure 1) in the first edition of *On the Origin of Species* (1859).

I am tempted to give one other case, the well-known one of the denudation of the Weald. Though it must be admitted that the denudation of the Weald has been a mere trifle, in comparison with that which has removed masses of our palæozoic strata, in parts ten thousand feet in thickness, as shown in Prof. Ramsay's masterly memoir on this subject. Yet it is an admirable lesson to stand on the North Downs and to look at the distant South Downs; for, remembering that at no great distance to the west the northern and southern escarpments meet and close, one can safely picture to oneself the great dome of rocks which must have covered up the Weald within so limited a period as since the latter part of the Chalk formation. The distance from the northern to the southern Downs is about 22 miles, and the thickness of the several formations is on an average about 1100 feet, as I am informed by Prof. Ramsay. But if, as some geologists suppose, a range of older rocks underlies the Weald, on the flanks of which the overlying sedimentary deposits might have accumulated in thinner masses than elsewhere, the above estimate would be erroneous; but this source of doubt probably would not greatly affect the estimate as applied to the western extremity of the district. If, then, we knew the rate at which the sea commonly wears away a line of cliff of any given height, we could measure the time requisite to have denuded the Weald. This, of course, cannot be done; but we may, in order to form some crude notion on the subject, assume that the sea would eat into cliffs 500 feet in height at the rate of one inch in a century. This will at first appear much too small an allowance; but it is the same as if we were to assume a cliff one yard in height to be eaten back along a whole line of coast at the rate of one yard in nearly every twenty-two years. I doubt whether any rock, even as soft as chalk, would yield at this rate excepting on the most exposed coasts; though no doubt the degradation of a lofty cliff would be more rapid from the breakage of the fallen fragments. On the other hand, I

Outline for whiteboard

- Project feedback
- HW #3 due today
- HW #2 comment?
- hosting – markers
- Evolution & Deep Time
- Darwin's Estimate & Update
- – Mention Kelvin
- Molecular Clock
- Size of the Biosphere
- Size of Humanity

do not believe that any line of coast, ten or twenty miles in length, ever suffers degradation at the same time along its whole indented length; and we must remember that almost all strata contain harder layers or nodules, which from long resisting attrition form a breakwater at the base. Hence, under ordinary circumstances, I conclude that for a cliff 500 feet in height, a denudation of one inch per century for the whole length would be an ample allowance. At this rate, on the above data, the denudation of the Weald must have required 306,662,400 years; or say three hundred million years.

—Charles Darwin, *Ch. 9 of On the Origin of Species*, 1859

The Weald is a region in Southeastern England that is a depression between two parallel chalk escarpments. Darwin reasoned that prior chalk deposits had eroded to form the modern Weald. This brought up a comparison to the White Cliffs of Dover, which are also a chalk formation.

Darwin estimated that seawater would "eat into cliffs" at a **rate of 1 inch per century**. He imagined the Weald as cliffs on their side (Fig. 3) and assumed erosion removes the same volume per time from the Weald as from the Cliffs. The rate of erosion (volume removal) is then

$$E = \frac{dV}{dt} = LH \frac{dW}{dt}.$$

On average, this rate is the same across the length of the cliffs (8 mi $\approx$ 13 km).

$$\frac{E}{L} = H \frac{dW}{dt} \approx 500 \text{ ft} \times 12 \text{ in/ft} \times \frac{1 \text{ in}}{100 \text{ yr}} \approx 60 \text{ in}^2/\text{yr}$$

A chalk layer about 1100 ft tall ($\approx 10^4$ in) was removed across the 22 mi ($\approx 10^6$ in) width of the Weald. If this material was removed at the same rate, then

$$T = V/E = \frac{V/L}{E/L} = \frac{H_{\text{weald}} \cdot W_{\text{weald}}}{60 \text{ in}^2/\text{yr}} \approx \frac{10^4 \times 10^6 \text{ in}^2}{60 \text{ in}^2/\text{yr}} \approx f \times 10^8 \text{ yr}$$

Darwin reported his estimate with much greater precision (306 662 400 yrs), but clearly indicated that it should be rounded due to the many approximations made.

2.1 An updated estimate

Darwin's estimate is very interesting, but also has weaknesses, including:

1. He guessed an erosion rate for the Cliffs of Dover.
2. Erosion from seaside cliffs is different than inland hilltops.
3. Erosion rates are not constant in space or time.
4. E.g., in upland settings, streams carry eroded material away.
5. Darwin omits important processes like uplift.¹

The major problem is Darwin's guess about erosion rates (#1). To see the problem, let's rewrite Darwin's calculation as a linear erosion rate (height loss/yr)

$$\approx \frac{f \times 10^2 \text{ m eroded}}{f \times 10^8 \text{ yr}} \times \frac{10^3 \text{ mm}}{1 \text{ m}} \approx 10^{-3} \text{ mm/yr}.$$

As shown in Fig. 4, measurements of inland erosion rates are often higher than this, especially in agricultural regions. Modern estimates for the Weald

¹A modern treatment uses partial differential equations to describe the effects erosion, stream carving, uplift, etc. on the evolution of landscapes. You can find an introduction in this [chapter](#) by J.D. Pelletier.



Figure 2: The White Cliffs of Dover, from [Wikipedia](#).

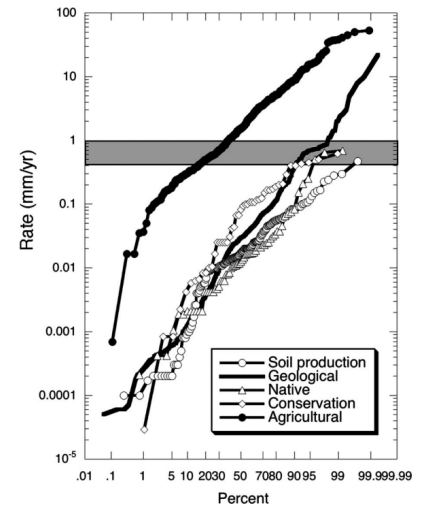


Figure 4: Distributions of modern erosion rate measurements from [Montgomery, PNAS 2007](#).

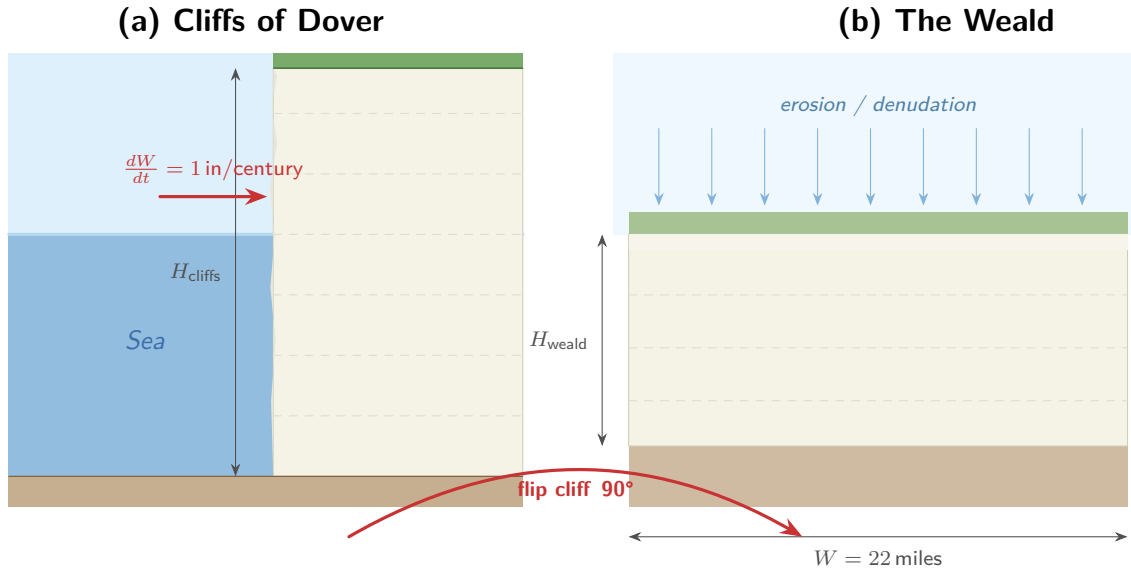


Figure 3: The essence of Darwin's calculation is the assumption of a constant rate of erosion per unit volume shared by the Weald and the Cliffs of Dover. Graphic made with help from Claude.

specifically (Jones, Geol. Soc. Spec. Pub. 1999) are about tenfold higher than Darwin's guess:

$$\approx \frac{20 \text{ m}}{10^6 \text{ yr}} \times \frac{10^3 \text{ mm}}{1 \text{ m}} = 2 \times 10^{-2} \text{ mm/yr.}$$

Using an erosion rate of $f \times 10^{-2} \text{ mm/yr}$ gives²

$$T \approx \frac{f \times 10^2 \text{ m} \times 10^3 \text{ mm} / \text{m}}{f \times 10^{-2} \text{ mm/yr}} \approx 10^7 \text{ years}$$

3 Age of the Earth – Kelvin's heat diffusion

Charles Lyell convinced many geologists to adopt a "uniformitarian" stance – assuming that the processes like erosion that take place today have taken place in the same way over Earth's history. Darwin applied a uniformitarian idea by using a contemporary erosion estimate to project backwards in time.

William Thomson (Lord Kelvin) objected that present-day geological rates could not safely be extrapolated over long periods of history. Kelvin took a different approach to Earth's age by **estimating the time it would take for Earth to cool**.

Heat transport in solids is a diffusive process.³ Treating the Earth as a conducting sphere with thermal diffusivity D_T , he applied Fourier's theory of heat transfer to estimate a cooling timescale.

We can try to reconstruct this estimate via dimensional analysis. We want t_{cooling} with dimensions of time $[T]$. We think that the thermal diffusivity D_T

²Darwin's assumed rate of $1 \text{ in/century}^{-1} \approx 0.25 \text{ mm yr}^{-1}$ falls between the two middle rows, giving $T \approx 300 \text{ Myr}$. Modern rates for lowland chalk landscapes are typically 10^{-2} – $10^{-1} \text{ mm yr}^{-1}$, implying timescales of 3–30 Myr.

³Fourier developed the theory of heat transfer in solids in 1822, well before Brown, Einstein, and Perrin.

Erosion rate E (mm yr ⁻¹)	Timescale T (yr)
10^{-3}	3×10^8
10^{-2}	3×10^7
10^{-1}	3×10^6
1	3×10^5

Table 1: Timescale $T = H/E$ for denudation of the Weald, assuming a uniform chalk thickness $H = 300 \text{ m}$.

$[L^2/T]$, the temperature gradient $\frac{dT}{dz}$ $[K/L]$, and the temperature of molten rock, T_{molten} $[K]$ all affect the result.

$$t_{\text{cooling}} \propto T_{\text{molten}}^{\alpha} D_T^{\beta} \left(\frac{dT}{dz} \right)^{\gamma}$$

Giving

$$T = K^{\alpha} (L^2/T)^{\beta} (K/L)^{\gamma} = K^{\alpha+\gamma} L^{2\beta-\gamma} T^{-\beta}$$

We need T^1 so, $\beta = -1$. We need the dimensions of L to cancel, which means $2\beta - \gamma = 0$ and $\gamma = -2$. Finally, we need the dimensions of K to cancel, so $\alpha + \gamma = 0$ and $\alpha = 2$. This gives

$$t_{\text{cooling}} \propto T_{\text{molten}}^2 D_T^{-1} \left(\frac{dT}{dz} \right)^{-2} = \frac{T_{\text{molten}}^2}{D_T \cdot (dT/dZ)^2}$$

The dimensionless constants can be derived with some calculus,⁴ which gives

$$t_{\text{cooling}} = \frac{T_{\text{molten}}^2}{\pi \cdot D_T \cdot (dT/dZ)^2}$$

Kelvin assumed a **molten rock temperature of** $T_{\text{molten}} \approx f \times 10^3 \text{ K}$ and recent measurements of **heat diffusion** $D_T \approx 10^{-6} \text{ m}^2/\text{s}$. From measurements in deep mines, he knew the temperature gradient near the surface $dT/dz \approx \frac{f \text{ K}}{100 \text{ m}}$.

$$t_{\text{cooling}} \approx \frac{(f \times 10^3)^2 \text{ K}^2}{\pi \cdot 10^{-6} \text{ m}^2/\text{s} \cdot \left(\frac{f \text{ K}}{100 \text{ m}} \right)^2} \approx \frac{10^7}{\pi \times 10^{-6} \times 10^{-3}} \approx f \times 10^{15} \text{ s}$$

$$f \times 10^{15} \text{ s} \times \frac{1 \text{ yr}}{\pi \times 10^7 \text{ s}} \approx 10^8 \text{ yr}$$

Kelvin's estimate posed problems for geologists and biologists, who felt that much more time was needed for the evolution of landscapes and life. It seemed very solid at the time, but Kelvin didn't know about radioactivity, which provides an ongoing source of heat to the Earth. For a nice summary of this history, see [Badash, Sci. Am, 1986](#).

4 Age of the Earth – from genomes

Here's our rationale: the Earth is older than life. We think bacterial life is very old. We'll try estimate the age of the bacteria to get a lower bound on the age of the Earth.

Thoughts about how to estimate the age of bacteria?

Since we live in the age of genomes, we'll estimate about the time it takes to turn one genome into another.

Two very diverged bacteria still share some identifiable genes that they got from their common ancestor, e.g., the β subunit of RNA polymerase (rpoB). By aligning these we can estimate number of amino acid substitutions. We also have empirical estimates of the rate of divergence (substitutions/site/year) from modern bacteria, so

$$\frac{\# \text{ substitutions}}{\text{N sites}} \times \frac{1}{\text{subs/site/yr}} \approx \text{years elapsed}$$

⁴See Rob Phillips forthcoming book "The Art of Estimation" which covers this topic. The original reference is William Thompson (Lord Kelvin) "On the Secular Cooling of the Earth," Trans. Royal Soc. Edinburgh, 1862.

The *rpoB* gene encodes a subunit of RNA polymerase. This gene is universally present in bacterial genomes, found in single copy, and slow-evolving. I aligned *rpoB* sequences from *E. coli* and *B. subtilis*,⁵. The aligned sequences have about 600 differences $\approx$ 1200 aligned amino acids ($\approx$ 60% identical).

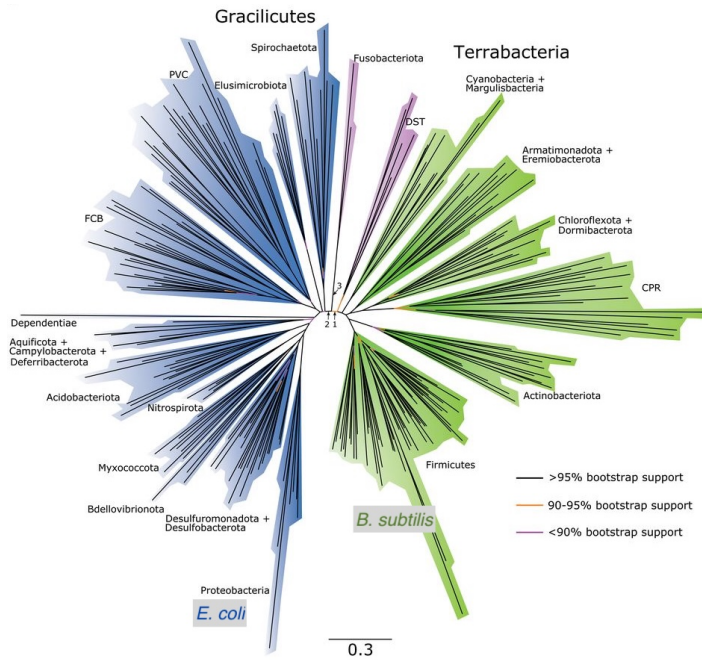


Figure 5: A rooted tree of bacterial life from [Coleman et al., Science 2021](#). The approximate positions of *E. coli* and *B. subtilis* are marked.

4.1 Poisson correction

Mention in class, but don't go into detail.

40% of the sites differ between *E. coli* and *B. subtilis* *rpoB*. But some of these sites may have mutated more than once, e.g.,

glutamate (E) $\rightarrow$ aspartate (D) $\rightarrow$ glutamine (Q)

as in Fig. 6. Also, **identical sites might have mutated and reverted**, e.g.,

lysine (K) $\rightarrow$ arginine (R) $\rightarrow$ lysine (K)

Since there are mutations we can't see, the fraction of non-synonymous sites $p = 0.4$ is different from the per-site mutation rate.

Since mutation is rare and random, mutations should follow a [Poisson distribution](#). The probability of seeing k mutations at a rate d is

$$P(k) = \frac{d^k e^{-d}}{k!}; \quad P(k=0) = e^{-d}; \quad P(k>0) = 1 - e^{-d}$$

Since we see one or more mutations at a frequency (rate) $p \approx 0.4$ and

$$1 - p = e^{-d} \implies d = -\ln(1 - p)$$

using $p = 0.4$ gives a Poisson corrected rate of $d \approx 0.5$.

These values are not so different and would not change the order of magnitude of our result.

⁵UniProt entries [P37870](#) and [P0A8V2](#) aligned with [MAFFT](#).

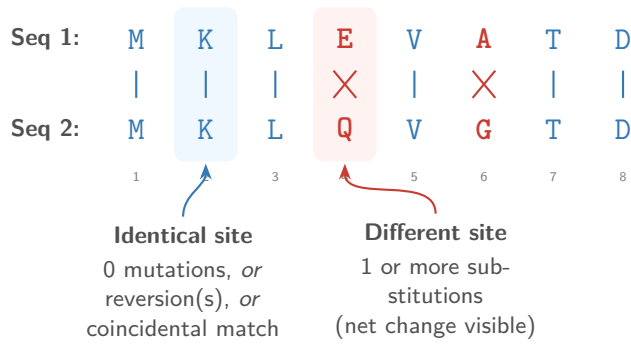


Figure 6: The Poisson correction estimates the *true* number of substitutions per site, accounting for multiple hits hidden beneath identical-looking positions. Made with the help of Claude.

4.2 Empirical mutation rate for proteins

This is the most challenging value to measure / estimate. To get an estimate, we use evolutionary splits that occurred at approximately known times. For bacteria, a common one is the *E. coli* / *Salmonella* split, as in [Doolittle et al. Science 1996](#).

Table 3. Average resemblances and divergence times by extrapolation.

	N*	ID† (%)	D‡	LCA§	LCA'	LCA''	LCA'''
Deuterostome-protostome	21	64 ± 10	36	750	656	675	675
Schizocoelome-pseudocoelome	9	64 ± 8	37	750	784	750	750
Fungi-animal	54	55 ± 8	52	1050	978	965	965
Plant-animal	33	57 ± 8	47	1000	1000	1000	1000
Protist-plant-animal-fungi	14	51 ± 10	59	1250	1236	1230	1230
Archaeobacteria-eukaryotes	9	39 ± 6	85	1800	1889	1700	1870
Eubacteria-eukaryotes	57	37 ± 9	96	2050	2080	1875	2156
<i>Bacilli-E. coli</i>	28	45 ± 9	75	(1500)	1610	1450	1523
<i>E. coli-Salmonella</i>	8	94 ± 6	6	(100)	(100)	(100)	(100)

*Number of enzyme sets compared. †Percent identity ± SD. ‡Distances, from Fig. 2C. §LCA, last common ancestor given as million years ago (LCA from Fig. 2C); LCA', average of Fig. 3, A and B; LCA'', after scaling (Fig. 3C); and LCA''', after correction for amino acid replacement constraints.

Figure 8: Estimates of protein sequence divergence rates from [Doolittle et al. Science1996](#)

Considering the *E. coli* / *Salmonella* split in Figure 8 suggests a 5% sequence divergence over 100 million years, giving a rate of

$$\frac{0.05 \text{ differences / site}}{10^8 \text{ years}} \approx 5 \times 10^{-10} \text{ mutations/site/year}$$

This agrees roughly with BioNumbers, which gives a textbook value of 10^{-9} amino acid substitutions/site/year [BNID [100358](#)].

4.3 Simple calculation

Use these values to make an estimate.

$$\frac{0.5 \text{ mutations/site}}{10^{-9} - 10^{10} \text{ mutations/site/yr}} \approx \frac{0.5}{10^{-9} - 10^{-10}} \approx 5 \times 10^8 - 5 \times 10^9 \text{ years}$$

While this is a wide spread, it's not 8000 years or 10 million years. We need more time than that. Of course our result depends on what rate we assume (as in the table below), but realistic rates imply that life is very old.

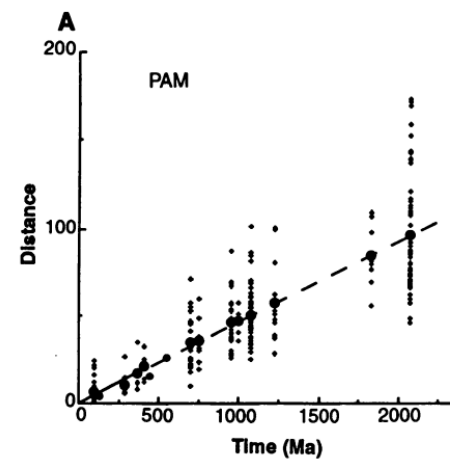


Figure 7: Divergence of protein sequences over time from [Doolittle et al. Science 1996](#). Notice that the Y axis units are distances calculated according to an amino acid similarity score (PAM). Nonetheless, notice that one can estimate a rate of divergence by plotting sequence differences against time for lineages where we have good estimates of the timing of their divergence, e.g., from fossils or other knowledge.

Rate μ (sub site ⁻¹ yr ⁻¹)	T (years)	Plausible?
10^{-9}	$f \times 10^8$	Too recent ($\sim$ Carboniferous)
10^{-10}	$f \times 10^9$	Consistent with Earth age
10^{-11}	$f \times 10^{10}$	Older than the universe

5 The size of the biosphere

Today we have an oxygenated atmosphere and a vibrant biosphere.

How big is the biosphere? How should we quantify this? **I think mass of carbon is a good way.**

When I'm walking, biking, or driving I see a lot of plants, even in NYC. So we'll estimate the mass of trees to get a sense of the scale of the biosphere.

Divide and conquer approach:

$$\boxed{\text{trees / area}} \times \boxed{\text{global forest area}} \times \boxed{\text{carbon mass / tree}}$$

5.1 Carbon mass of a single tree

Model a tree as a cylinder of height $h = 10$ m and a $1/2$ m diameter (radius $r = 0.5$ m)

$$V_{\text{tree}} \approx \pi r^2 h \approx 3 \times 0.1 \times 10 \approx f \text{ m}^3.$$

Dry wood is less dense than water (it floats) and has a density of $\rho \approx 0.6$ kg / L = $\boxed{600 \text{ kg m}^{-3}}$. So the above-ground mass is

$$\approx f \text{ m}^3 \times 600 \text{ kg m}^{-3} \approx 1 - 2 \times 10^3 \text{ kg dry wood}$$

Dry wood is about 50% carbon, while the trunk is about 1/2 the trees mass (roots $\approx 20 - 30\%$, canopy $\approx 10 - 20\%$). So the carbon mass of a tree is

$$\approx 10^3 \text{ kg dry trunk} \times \frac{2 \text{ kg dry tree}}{\text{kg dry trunk}} \times \frac{1 \text{ kg C}}{2 \text{ kg dry wood}} \approx 10^3 \text{ kg C / tree}$$

5.2 Tree density from spacing

How far apart are trees?

I think forest trees are spaced a f m apart. But a lot of these trees are small. Let's take a spacing of 10 m. Imagine a square km with a grid on it. It has an area of 10^6 m^2 and $(1000/10)^2 \approx 10^4$ trees.

$$\rho_{\text{trees}} \approx \frac{10^4 \text{ trees}}{\text{km}^2}$$

5.3 Putting it all together

About 1/3 of the land surface is forested. With $\boxed{A_{\text{land}} \approx 1.5 \times 10^8 \text{ km}^2}$, we get a total forest carbon mass of

$$1.5 \times 10^8 \text{ km}^2 \text{ land} \times \frac{1 \text{ km forest}}{3 \text{ km land}} \times \frac{10^4 \text{ trees}}{\text{km}^2} \times 10^3 \text{ kg C / tree} \approx 5 \times 10^{14} \text{ kg C}$$

Our World
in Data

Global biomass: 546 billion tonnes of carbon

Category	Biomass (billion tonnes carbon)	% of total biomass
Plants	450	82.4%
Animals	96	17.6%
Arthropods	42	7.7%
Fish	29	5.3%
Other animals	25	4.6%

Animal biomass breakdown (96 billion tonnes carbon):

Group	Biomass (billion tonnes carbon)	% of animal biomass
Arthropods	42	42%
Fish	29	29%
Other animals	25	29%

Other animals breakdown (25 billion tonnes carbon):

Group	Biomass (billion tonnes carbon)	% of other animals
Amphibians	0.1	0.4%
Reptiles	0.1	0.4%
Birds	0.1	0.4%
Mammals	0.1	0.4%
Other	0.1	0.4%

Plants breakdown (450 billion tonnes carbon):

Group	Biomass (billion tonnes carbon)	% of plants
Algae	0.1	0.02%
Plants	0.1	0.02%
Fungi	0.1	0.02%
Other	0.1	0.02%

Other animals breakdown (25 billion tonnes carbon):

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Amphibians	0.1	0.4%
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This is about 500 Gigatonnes / Petagrams⁶ of carbon or 5×10^{17} g C. This is in good agreement with the 4.5×10^{17} g C estimated for all plant biomass by Bar-On et al. 2018, or $\approx 80\%$ of the total C mass of the biosphere ($\approx 5.5 \times 10^{17}$ g C). A very vibrant biosphere indeed.

Approach

$$\boxed{N_{\text{ppl}}} \times \boxed{M_{\text{person}}} \times \boxed{F_{\text{water}}} \times \boxed{F_C}$$

where

$N_{\text{ppi}} \approx 8 \times 10^9$

$M_{\text{person}} \approx 70 \text{ kg}$

$F_{\text{water}} \approx 0.7$

$F_C \approx 0.5$

6.1 A generous over-estimate

$$\approx 10^{10} \text{ ppl} \times 10^2 \text{ kg/person} \times (1-0.7) \frac{\text{kg dry person}}{\text{kg wet person}} \times 0.5 \frac{\text{kg C}}{\text{kg dry person}} \approx 10^{11} \text{ kg C}$$

Or **5000** times less carbon in people than in trees.

The human population is not all adults, and adults don't usually weigh $100kg$. Including children and women gives a mean body mass closer to $50kg$. We should also use the actual population of 8×10^9 people:

$$50 \text{ kg} \times 8 \times 10^9 \times 0.15 \text{ kg C/kg} \approx 6 \times 10^{10} \text{ kg}$$

So **human bodies make up $\approx 0.01\%$ of the biosphere**. Despite this, we make quite a mark. But maybe we don't have to?

$${}^6_1\text{Gt} = 1\text{Pg} = 10^{15}g$$